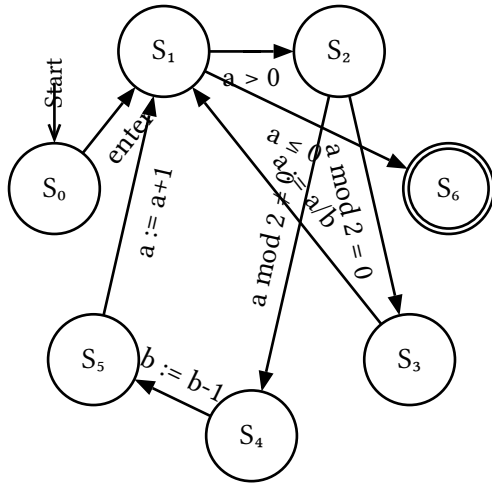


Abstract Interpretation I

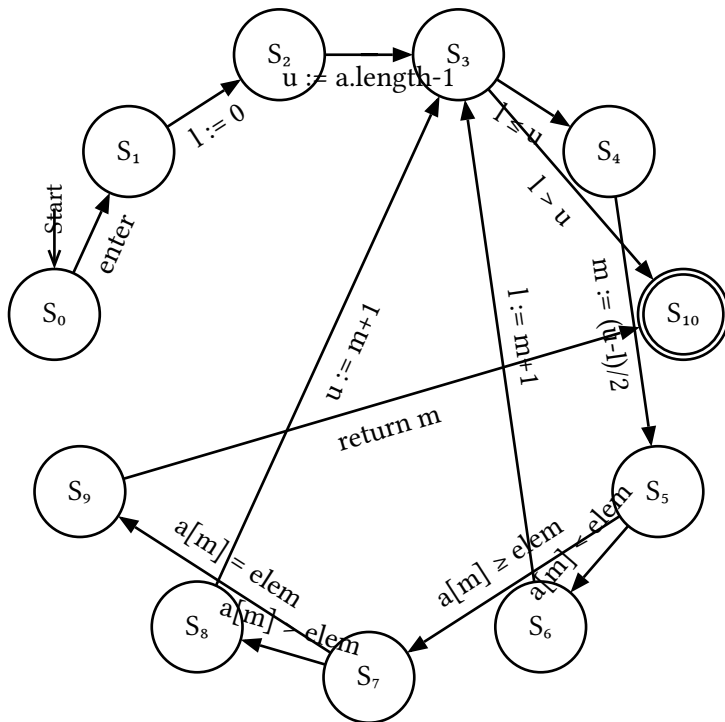
Building CFG

Given the following programs

Silly Program



Binary Search



Program Code

Silly Program Code

```
sillyprogram(a,b) =  
  while (a > 0) do  
    if a mod 2 = 0 then  
      a := a/b  
    else  
      b := b-1;  
      a := a+1
```

```

endif
done

```

Binary Search Code

```

binarysearch(a, elem) =
  l := 0;
  u := a.length-1;
  while (l <= u) do
    m := (u-l)/2;
    if (a[m] < elem) then
      l := m+1
    else
      if (a[m]>elem) then
        u := m+1
      else
        m;
      endif;
    endif
  done;
-1

```

Defining CFG

Define the function that builds a CFG, from a program P. To make the definition easier, we advise that the function also takes as inputs ip and rp which will be the entry and return control points of the resulting CFG. At last, we will define $CFG(P, ip, rp)$; it should depend on the grammar that defines the instructions of the language.

Value Equations for Silly Program

- $V(A) = T \cup V(C)[a := \frac{a}{b}] \cup V(E)[a := a + 1]$
- $V(B) = V(A) \cap [a > 0]$
- $V(C) = V(B) \cap [a \bmod 2 = 0]$
- $V(D) = V(B) \cap [a \bmod 2 \neq 0]$
- $V(E) = V(D)[b := b - 1]$
- $V(F) = V(A) \cap [a \leq 0]$

Development

- $T = \{a : [5], b : [3]\}$
- $V(E) = \{a : [5], b : [2]\}$
- $V(A) = V(E)[a := a + 1] \cup T$
- $V(A) = \{a : [4], b : [2]\} \cup \{a : [5], b : [3]\}$
- $V(A) = \{a : [4, 5], b : [2, 3]\}$
- $V(B) = V(A) \cap [a > 0] = \{a : [4, 5], b : [2, 3]\}$
 - For $a = 4 : 4 \bmod 2 = 0$ (even)
 - For $a = 5 : 5 \bmod 2 = 1 \neq 0$ (odd)
- $V(C) = V(B) \cap [a \bmod 2 = 0] = \{a : [4], b : [2, 3]\}$
- $V(D) = V(B) \cap [a \bmod 2 \neq 0] = \{a : [5], b : [2, 3]\}$
- $V(E) = V(D)[b := b - 1] = \{a : [5], b : [1, 2]\}$

But we're given $V(E) = \{a : [5], b : [2]\}$, so:

$V(D) = \{a: [5], b: [3]\}$ to get $V(E) = \{a: [5], b: [2]\}$

Therefore:

- $V(B) \supseteq \{a: [5], b: [3]\}$
- Since $V(B) = V(A) \text{ inter } [a > 0]$, we need $V(A)$ to contain $\{a: [5], b: [3]\}$
- Since $a = 5 > 0$, this constraint is satisfied

From $V(C) = V(B) \text{ inter } [a \bmod 2 = 0]$:

- For $a = 5: 5 \bmod 2 = 1 \neq 0$, so $V(C)$ does not contain $\{a: [5], b: [3]\}$
- We need values where $a \bmod 2 = 0$

Let's reconsider $V(A)$:

- $V(A) = T \text{ union } V(C)[a := a / b] \text{ union } V(E)[a := a + 1]$
- $T = \{a: [5], b: [3]\}$
- $V(E)[a := a + 1] = \{a: [4], b: [2]\}$ (since $4 + 1 = 5$)
- $V(C)[a := a / b]$: Need to find values where $a/b = 5$ or $a/b = 4$

If $V(C)[a := a / b] = \{a: [4], b: [2]\}$, then $V(C) = \{a: [8], b: [2]\}$ (since $8/2 = 4$) If $V(C)[a := a / b] = \{a: [5], b: [3]\}$, then $V(C) = \{a: [15], b: [3]\}$ (since $15/3 = 5$)

Therefore: $V(A) = \{a: [5], b: [3]\} \text{ union } \{a: [8], b: [2]\} \text{ union } \{a: [4], b: [2]\} = \{a: [4, 5, 8], b: [2, 3]\}$

Verification:

- $V(B) = V(A) \text{ inter } [a > 0] = \{a: [4, 5, 8], b: [2, 3]\}$
- $V(C) = V(B) \text{ inter } [a \bmod 2 = 0] = \{a: [4, 8], b: [2, 3]\}$
- $V(D) = V(B) \text{ inter } [a \bmod 2 \neq 0] = \{a: [5], b: [2, 3]\}$
- $V(E) = V(D)[b := b - 1] = \{a: [5], b: [1, 2]\}$

Still not matching exactly. Let's try with more precise intervals:

$V(A) = \{a: [4, 5], b: [2, 3]\}$

1.2

$\text{CFG}(P, ip, rp) = \text{CFG}(I_1; I_2; \dots; I_n, \top, f)$

- $\text{CFG}(x := e, ip_0, rp_0) = \text{automaton here}$
- $\text{CFG}(\text{if cond then } op_1 \text{ else } op_2, ip_0, rp_0) = \text{automaton here}$
- $\text{CFG}(op_1, ip_{op_1}, rp_0) \cup \text{CFG}(op_2, ip_{op_2}, rp_0)$
- $\text{CFG}(\text{while cond then } op_1, ip_0, rp_0) = \text{automaton here}$
- $\text{CFG}(op_1, ip_{op_1}, rp_0) \cup \underbrace{\text{CFG}\left(\underbrace{\emptyset}_{\text{pass}}, ip_0, rp_0\right)}_{\text{cond is false}}$

So $\text{CFG}(I_1; I_2; \dots; I_n, \top, f) = \text{CFG}(I_1, \top, rp_{I_1}) \cup \text{CFG}(I_2, rp_{I_1}, rp(I_2)) \cup \text{CFG}(I_3, rp_{I_2}, rp(I_3))$

- $V(A) = T$
- $V(B) = V(A)[a := 5]$
- $V(C) = V(B)[b := 3] \cup V(E)[a := \frac{a}{b}] \cup V(G)[a := a + 1]$
- $V(D) = V(C) \wedge [a > 0]$
- $V(E) = V(D) \wedge [a \% 2 = 0]$
- $V(F) = V(D) \wedge [a \% 2 \neq 0]$
- $V(G) = V(F)[b := b - 1]$
- $V(H) = V(C) \wedge [a \leq 0]$

Iterations

We should stop if we get some similar values !

0. $V(C) = \{a : [5], b : [3]\}$
1. $V(C) = \{a : [5, 6], b : [2, 3]\}$ because we have $a := a + 1$ and $b := b - 1$
2. $V(C) = \{a : [2, 6], b : [1, 3]\}$
3. $V(C) = \{a : [0, 6], b : [0, 3]\}$
4. $V(C) = \{a : [0, 6], b : [-1, 3]\}$
5. $V(C) = \{a : [-6, 6], b : [-2, 3]\}$
- $\vdots$
- ∞ . $V(C) = \{a : [-6, 6], b : [-\infty, 3]\}$

Abstraction

0. $V(C) = \{a : > 0, b : > 0\}$
1. $V(C) = \{a : \geq 0, b : \geq 0\}$
2. $V(C) = \{a : \geq 0, b : \top\}$
3. $V(C) = \{a : \top, b : \top\}$
4. $V(C) = \{a : \top, b : \top\}$
5. Stop.... However, we can't say for which value the program will finish...